

Diagnostic accuracy of optical coherence tomography for early oral cancer detection: A systematic review

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ABSTRACT

Purpose: This systematic review evaluated the diagnostic performance of optical coherence tomography (OCT) for the early detection of oral cancer, particularly emphasizing sensitivity, specificity, and overall diagnostic accuracy.

Materials and Methods: A comprehensive literature search was conducted across PubMed, ScienceDirect, and Wiley Online Library covering publications from 2015 to 2025, supplemented by manual hand-searching of relevant references. Studies were selected using the Population, Intervention, Comparison, Outcome, Study Design (PICOS) framework. Eligible studies evaluated the diagnostic accuracy of OCT using histopathology as the reference standard. Risk of bias was assessed using the QUADAS-2 tool, and the review methodology followed PRISMA 2020 and PRISMA-DTA guidelines. The review protocol was registered in PROSPERO (CRD420251112254).

Results: Seven studies met the predefined inclusion criteria. OCT demonstrated high diagnostic performance (sensitivity: 81.5%-100%, specificity: 68.8%-100%), with diagnostic accuracy reaching up to 100% in certain settings. Studies that incorporated machine learning approaches, including convolutional neural networks and support vector machines, consistently achieved superior diagnostic performance compared with conventional OCT interpretation alone. Overall methodological quality was generally low, with several studies exhibiting moderate to high risk of bias in specific domains.

Conclusion: OCT, particularly when augmented with artificial intelligence, demonstrates high diagnostic accuracy as a non-invasive imaging modality for the early detection of oral cancer. Its capability to identify dysplastic and malignant changes at the microstructural level offers meaningful diagnostic advantages over conventional examination methods. Nevertheless, larger-scale studies employing standardized protocols are required to confirm its clinical utility and support integration into routine oral cancer screening and diagnostic pathways. (*Imaging Sci Dent* 2026; 56: 1-10)

KEY WORDS: Diagnosis; Tomography, Optical Coherence; Oral Cancer

Introduction

Oral cancer remains a major global health challenge, contributing substantially to morbidity and mortality worldwide. In 2020, approximately 377,713 new cases and 177,757 deaths were attributed to cancers of the oral cavi-

ty globally. The disease burden is particularly pronounced in regions with high consumption of tobacco, alcohol, and areca nut, especially in South and Southeast Asia.^{1,2} Prognosis is strongly dependent on the stage at diagnosis; however, many cases of oral cancer are still identified at advanced stages, markedly reducing the likelihood of favorable outcomes. The five-year survival rate remains relatively low, at approximately 50-60%, highlighting the critical importance of early detection, particularly among high-risk populations. Delayed or inaccurate diagnosis of early oral malignancies continues to pose a serious clinical challenge, especially when malignant lesions clinically resemble com-

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mon dental infections. The World Health Organization and other international health agencies have therefore emphasized the need for improved early detection strategies and more effective diagnostic technologies to reduce the global burden of oral cancer.³⁻⁵

Current diagnostic approaches for oral lesions have important limitations. Visual oral examination (VOE), which serves as the conventional initial screening method, is highly dependent on clinician experience and may fail to detect subtle premalignant changes, including homogeneous leukoplakia and erythroplakia. Histopathological assessment via tissue biopsy remains the diagnostic gold standard; however, it is invasive, time-consuming, and impractical for large-scale or repeated screening. Patients may be reluctant to undergo repeated biopsies, particularly in cases involving extensive or multifocal lesions, and biopsy sampling itself is subject to sampling error and interpretative variability, potentially leading to false-negative results. Less invasive alternatives, such as exfoliative cytology or brush biopsy, generally demonstrate lower sensitivity and frequently produce indeterminate or non-diagnostic findings. Collectively, these limitations highlight the need for a rapid, non-invasive adjunctive diagnostic tool capable of facilitating timely identification of early-stage oral malignancy. Thus, there is a clear and pressing demand for reliable, non-invasive methods to support screening and triage of oral mucosal lesions.^{6,7}

In this context, OCT has emerged as a promising index test. OCT is a non-invasive imaging modality that generates real-time, high-resolution cross-sectional images of the oral mucosa, enabling visualization of both epithelial and subepithelial microstructures. By employing low-coherence near-infrared light rather than acoustic waves, OCT achieves axial resolutions of approximately 10-20 μm , allowing tissue visualization comparable to low-magnification histology. Advances in Fourier-domain OCT, including spectral-domain OCT and swept-source OCT, have further improved imaging speed and sensitivity relative to earlier time-domain OCT systems. Operating at wavelengths typically ranging from 800 to 1310 nm, OCT provides an optimal balance between tissue penetration and image resolution, enabling non-invasive, real-time assessment of subsurface microstructural alterations.⁸⁻¹⁰

The diagnostic advantages of OCT over conventional approaches are considerable. The technique is non-contact, non-ionizing, and enables rapid image acquisition without the use of dyes or contrast agents. Its high spatial resolution permits detailed visualization of microarchitectural features that are critical for identifying dysplasia and

early-stage carcinoma, including epithelial thickness alterations, surface keratinization patterns, basement membrane continuity, and early signs of invasion into the lamina propria. Although OCT has a relatively limited penetration depth of approximately 1-3 mm, this range is sufficient to assess the full thickness of the oral mucosa and superficial submucosa, making it particularly well suited for detecting early-stage lesions that may be overlooked during routine visual examination.^{11,12}

For OCT to function effectively as an index test in clinical practice, it is essential to define minimally acceptable diagnostic performance thresholds. Prior diagnostic research has suggested sensitivity and specificity values of at least 80% as benchmarks for clinical utility in early cancer detection, as performance below this level is unlikely to provide substantial benefit over conventional examination alone. In comparative evaluations, a minimum improvement of approximately 10% in diagnostic accuracy relative to existing adjunctive techniques, such as toluidine blue staining or autofluorescence imaging, has been proposed as clinically meaningful. Establishing these performance thresholds is critical for interpreting the potential role of OCT within diagnostic pathways and for evaluating the strength, quality, and applicability of the existing evidence base.¹⁰

Given these considerations, the aim of this systematic review is to evaluate the diagnostic performance of optical coherence tomography (OCT), as measured by sensitivity, specificity, and accuracy, in the detection of oral cancer. In addition, this review seeks to clarify the potential contribution of OCT to improved early detection of oral malignancies and its role in supporting better clinical outcomes for affected patients.

Materials and Methods

The conduct and reporting of this systematic review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement and the PRISMA Diagnostic Test Accuracy (PRISMA-DTA) guidelines. The protocol for this systematic review was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO; CRD 420251112254).

The scope of this systematic review was structured using the Population, Intervention, Comparison, Outcome, Study Design (PICOS) framework. The population consisted of patients presenting with oral mucosal lesions clinically suspicious for precancerous changes or early-stage oral

Table 1. PICOS structure of the study

Population	Patients presenting with oral mucosal lesions suspicious for precancerous changes or early-stage oral squamous cell carcinoma (OSCC)
Intervention	Optical coherence tomography (OCT), including conventional systems and/or OCT enhanced with artificial intelligence (AI)-based analysis
Comparison	Reference standard diagnostic methods, primarily histopathological examination with or without adjunctive clinical assessment
Outcome	Diagnostic accuracy of OCT, assessed by sensitivity, specificity, and/or overall accuracy
Study design	Human studies, including <i>in vivo</i> or <i>ex vivo</i> studies

squamous cell carcinoma (OSCC). The intervention was OCT, including both conventional OCT systems and OCT enhanced with artificial intelligence-based analytical approaches. The comparison was the reference standard of histopathological examination, with or without adjunctive clinical assessment. The primary outcomes were diagnostic accuracy measures, specifically sensitivity, specificity, and/or overall diagnostic accuracy, of OCT for detecting oral precancerous lesions and early-stage OSCC. Eligible study designs included human studies conducted *in vivo* or *ex vivo*, as summarized in Table 1.

Eligibility criteria

This systematic review focused on studies evaluating the diagnostic performance of OCT for the detection of early-stage oral cancer and potentially malignant oral disorders. Eligible studies were required to: 1) employ OCT as the index test and provide sufficient data to calculate or directly report diagnostic performance metrics, including sensitivity, specificity, and/or overall diagnostic accuracy; 2) be primary diagnostic accuracy studies, encompassing prospective designs as well as intraoperative, *in vivo*, and *ex vivo* applications; and 3) be peer-reviewed full-text articles published in English between January 2015 and May 2025.

Studies were excluded if they did not present original data on the diagnostic performance of OCT or if they constituted secondary analyses, including narrative reviews, systematic reviews, or meta-analyses. Case reports, editorials, expert opinion articles, and conference abstracts were also excluded, as these publication types do not provide extractable diagnostic accuracy data.

Search strategy and data collection

To identify relevant literature, a comprehensive search was conducted across multiple electronic databases, including PubMed, ScienceDirect, Wiley Online Library, and

Google Scholar, which was used for grey literature retrieval and hand searching. The search was restricted to studies published between 2015 and 2025, a timeframe deliberately selected to capture the most recent and methodologically relevant evidence on the diagnostic performance of OCT for early oral cancer detection.

The search strategy combined predefined keywords and Medical Subject Headings (MeSH) terms to identify eligible studies. The terms used included “Optical Coherence Tomography,” “oral cancer,” “oral squamous cell carcinoma,” and “early diagnosis.” In addition, hand searches were conducted of well-established publications addressing the diagnostic performance of OCT in early oral cancer detection. Boolean operators, including AND and OR, were applied to optimize retrieval. A detailed summary of the search strategies and the total number of records retrieved is presented in Table 2.

The study selection process was performed in two sequential stages. First, all records retrieved from the electronic searches were imported into a reference management system, and duplicate entries were removed. Titles and abstracts were independently screened by two reviewers (MZW and FUAR) to exclude studies that were clearly irrelevant. In the second stage, the full texts of potentially eligible articles were retrieved and assessed in detail against the predefined inclusion and exclusion criteria. Discrepancies were resolved through discussion and, when necessary, consultation with a third reviewer (NEN). The complete selection process, including the number of records at each stage, reasons for exclusion, and the final set of included studies, is depicted in the PRISMA flow diagram (Fig. 1).¹³

Data extraction was conducted independently and in duplicate by two reviewers using a piloted, self-developed standardized data extraction form. Extracted data included study identifiers (first author, publication year, and country), study design, OCT system type and analytical approach, reference standard, and reported diagnostic

Table 2. Custom search strategy for each database

Databases	Search strategy used
PubMed	((("mouth neoplasms" [MeSH Terms] OR ("mouth" [All Fields] AND "neoplasms" [All Fields]) OR "mouth neoplasms" [All Fields] OR ("oral" [All Fields] AND "cancer" [All Fields]) OR "oral cancer" [All Fields] OR ("squamous cell carcinoma of head and neck" [MeSH Terms] OR ("squamous" [All Fields] AND "cell" [All Fields] AND "carcinoma" [All Fields] AND "head" [All Fields] AND "neck" [All Fields]) OR "squamous cell carcinoma of head and neck" [All Fields] OR ("oral" [All Fields] AND "squamous" [All Fields] AND "cell" [All Fields] AND "carcinoma" [All Fields]) OR "oral squamous cell carcinoma" [All Fields])) AND ("tomography, optical coherence" [MeSH Terms] OR ("tomography" [All Fields] AND "optical" [All Fields] AND "coherence" [All Fields]) OR "optical coherence tomography" [All Fields] OR ("optical" [All Fields] AND "coherence" [All Fields] AND "tomography" [All Fields])))
ScienceDirect	Optical coherence tomography AND oral cancer AND early diagnosis
Wiley Online Library	Optical coherence tomography AND oral cancer AND early diagnosis

outcomes (sensitivity, specificity, and/or accuracy). A third reviewer verified the extracted data for accuracy, and disagreements were resolved by consensus. When necessary, study authors were contacted to obtain clarification or missing information.

Assessing risk of bias

The methodological quality and risk of bias of the included diagnostic accuracy studies were assessed using the Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool. This instrument evaluates risk of bias and concerns regarding applicability across four domains: patient selection, index test, reference standard, and flow and timing. Each domain was assessed using signaling questions, with judgments categorized as low risk, high risk, or unclear.

Two reviewers independently applied the QUADAS-2 tool to all included studies. Any disagreements were resolved through discussion and, when necessary, consultation with a third reviewer to reach consensus. In parallel with risk of bias assessment, concerns regarding applicability to the review question were evaluated across the same four domains. Domain-level and overall judgments for each study are reported narratively and summarized in tabular form in the Results section, supporting transparency and reproducibility.

This structured assessment enabled identification of key methodological limitations, including potential selection bias, absence of blinding in the interpretation of index tests or reference standards, and incomplete reporting of participant flow. These quality considerations were subsequently incorporated into the interpretation of the diagnostic accuracy findings.

Synthesis of the summary measures

To synthesize the diagnostic performance of OCT, variations in target condition definitions were considered, with primary emphasis on early-stage oral cancer and precancerous lesions confirmed by histopathology. Variability in test positivity thresholds was addressed through narrative synthesis of diagnostic performance across reported cutoff points, with attention to clinical relevance. Indeterminate results were handled separately or excluded when appropriate to preserve analytical validity. In studies involving multiple readers, diagnostic outcomes were compared across different levels of expertise to assess potential reader-dependent effects on OCT performance.

For comparative synthesis, studies were grouped according to whether OCT was applied alone or in conjunction with artificial intelligence-based analytical techniques, highlighting the emerging role of artificial intelligence (AI) in improving diagnostic performance. This approach enabled a comprehensive evaluation of OCT across both conventional and AI-augmented implementations. Consistent use of histopathology as the reference standard across studies provided a robust basis for comparison, reinforcing the reliability of the synthesized diagnostic accuracy estimates. Overall, this synthesis underscores the evolving role of OCT, particularly when integrated with machine learning, as a promising non-invasive modality for early oral cancer detection.

Results

Study selection

The initial database search yielded 240 records, including PubMed (137 articles), ScienceDirect (94 articles), and Wiley Online Library (9 articles). An additional three records were identified through website searches. After

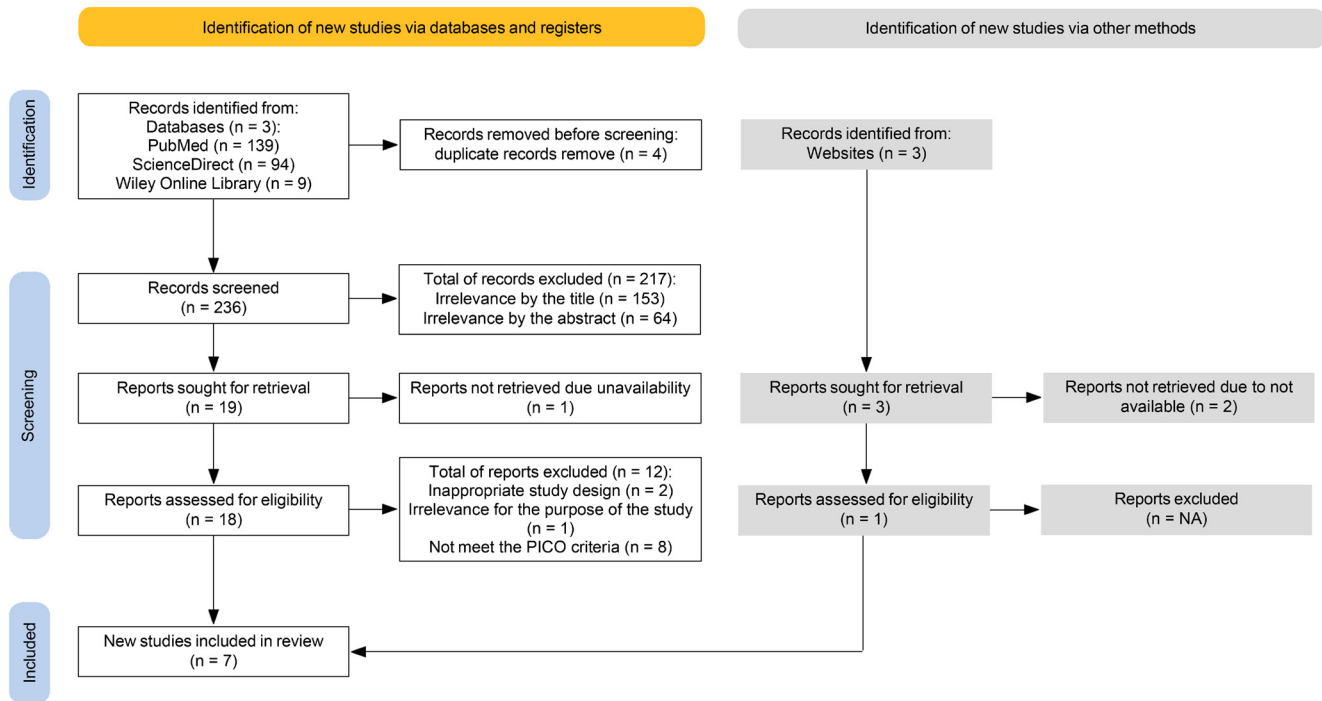


Fig. 1. PRISMA flow chart of the selection process.

removal of four duplicate records, 236 unique records remained for screening. Based on title and abstract screening, 217 records were excluded because they were not relevant to the review question, leaving 19 full-text articles for further evaluation.

Of the 19 full-text articles, one report could not be retrieved, and 18 articles were assessed for eligibility. Twelve studies were excluded because they did not meet the PICOS criteria, resulting in six eligible studies. Among the three records identified through website searches, two could not be retrieved, and one was assessed at the full-text level but subsequently excluded. Ultimately, six studies from the database search and one eligible study from the website search met all inclusion criteria, yielding a final total of seven studies included in the systematic review.¹⁴⁻²⁰

The complete study selection process, including the reasons for exclusion at each stage, is illustrated in the PRISMA flow diagram (Fig. 1).¹³

Risk of bias

The risk of bias across the included studies was assessed using the QUADAS-2 tool (Figs. 2 and 3). Overall, most assessed domains demonstrated a consistently low risk of bias, while the overall methodological quality of the included studies ranged from moderate to high.

The primary concerns were identified in the patient se-

lection domain, where non-consecutive sampling methods or insufficient reporting of recruitment procedures raised potential concerns regarding representativeness. Despite these issues, most studies (6 of 7) were judged to be at low risk of bias in this domain, indicating generally appropriate recruitment strategies. One study was rated at high risk due to potential selection bias.¹⁹ In the index test domain, four studies were rated at low risk, indicating that these studies adequately addressed the methodological requirements for the index test, including blinding to the reference standard and clear reporting of diagnostic thresholds. However, three studies raised concerns. One study was rated at high risk¹⁸ due to significant methodological issues, while two studies were rated as some concern^{16,19} due to insufficient details regarding blinding and transparency in the application of the index test. These issues emphasize the need for clearer reporting and better methodological rigor in future studies. The reference standard domain was generally robust, with six studies judged at low risk of bias. However, one study raised high risk¹⁹ due to incomplete descriptions of histopathological assessment procedures. Regarding flow and timing (Domain 4), 5 studies demonstrated low risk of bias, while 2 studies were rated at high risk because of incomplete follow-up or inappropriate intervals between the index test and reference standard.^{15,19}

Taken together, the overall quality of evidence was con-

	Risk of bias domains			
	D1	D2	D3	D4
Hamdoon et al. ¹⁴ 2016	+	+	+	+
Heidari et al. ¹⁵ 2019	-	+	+	X
Jerjes et al. ¹⁶ 2019	+	-	+	+
Sunny et al. ¹⁷ 2019	+	+	+	+
James et al. ¹⁸ 2021	+	X	+	+
Yang et al. ¹⁹ 2021	X	-	X	X
Yang et al. ²⁰ 2023	+	+	+	+

Domains:
D1: Patient selection
D2: Index test
D3: Reference standard
D4: Flow and timing

Judgement
X High
- Unclear
+ Low

Fig. 2. Risk of bias evaluation using QUADAS-2.

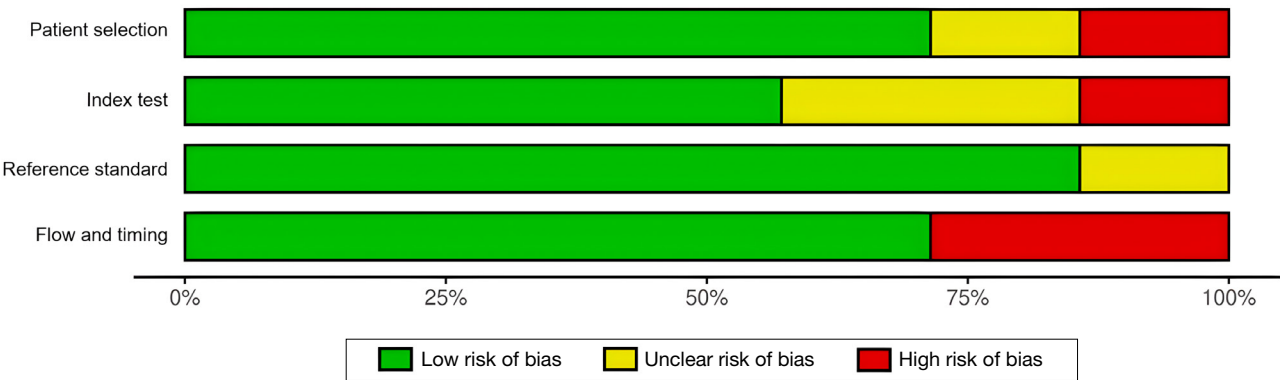


Fig. 3. Risk of bias evaluation using QUADAS-2.

sidered acceptable, with a predominance of studies demonstrating low risk of bias, supporting the methodological reliability of OCT diagnostic performance. Nevertheless, the presence of unclear and high-risk judgments highlights the need for further well-designed prospective studies with standardized reporting and rigorous methodological control.

Study characteristics

The included studies were conducted across diverse geographic settings, including 2 studies from the United Kingdom,^{14,16} 1 from the United States,¹⁵ 2 from India,^{17,18} and 2 from China.^{19,20} Sample sizes varied considerably, ranging from small exploratory studies^{17,19} involving 14 patients to larger clinical cohorts comprising up to 232 patients.¹⁸ All included studies evaluated patients presenting with potentially malignant oral disorders or early-stage oral

squamous cell carcinoma. Assessments were performed either intraoperatively^{17,18} or in clinical and *ex vivo* diagnostic contexts.^{14-16,19,20} Clinical settings varied from operating theaters to diagnostic laboratories, reflecting both *in vivo* and *ex vivo* applications of OCT. With respect to study design, most investigations were prospective diagnostic accuracy studies, including both intraoperative assessments and *ex vivo* imaging analyses. OCT served as the index test in all studies, with several incorporating advanced computational approaches, such as convolutional neural networks (CNN), support vector machines (SVM), artificial neural networks (ANN), or texture feature analysis, to enhance diagnostic performance. Histopathology was used as the reference standard in all studies, with one study additionally incorporating clinical examination. Diagnostic outcomes were consistently reported in terms of sensitivity, specificity, and/or overall ac-

curacy. Reported sensitivity values ranged from 81.5% to 100%, specificity from 68.8% to 100%, and overall accuracy reached up to 100% in selected intraoperative settings.¹⁷

Results of the studies

Across the included studies, OCT, with or without AI enhancement, demonstrated consistently high sensitivity, specificity, and diagnostic accuracy for early oral cancer detection when compared with histopathology as the reference standard.

Hamdoon et al.¹⁴ conducted an *ex vivo* study involving 20 patients and reported a sensitivity of 81.5% and specificity of 87%, resulting in an overall diagnostic accuracy of 84%. Heidari et al.¹⁵ performed an *in vivo* diagnostic accuracy study in 28 patients, in which OCT achieved a sensitivity of 95% and specificity of 100%, compared with sensitivity of 89% and specificity of 100% for conventional clinical examination. Jerjes et al.¹⁶ evaluated 60 patients and demonstrated that OCT-based epithelial thickness measurements yielded a sensitivity of 92.2% and specificity of 94.2%, supporting its utility in distinguishing dysplastic from non-dysplastic lesions. Sunny et al.¹⁷ assessed 14 patients intraoperatively and reported perfect diagnostic performance for malignancy detection, with sensitivity, specificity, and accuracy all reaching 100% (95% confidence interval [CI] for sensitivity: 85.2-100%; for specificity: 85.8-100%). For dysplasia detection, sensitivity was 92.5% (95% CI: 84.4-97.2), specificity was 68.8% (95% CI: 41.3-88.9), and accuracy was 88.5%. James et al.¹⁸ evaluated 232 patients and compared OCT alone with AI-assisted approaches. OCT combined with simple machine learning achieved a sensitivity of 93% (95% CI: 82-98), specificity of 74% (95% CI: 65-82), and overall accuracy of 85%. Use of an ANN-SVM hybrid model further improved diagnostic performance, yielding a sensitivity of 86% and specificity of 81%. Yang et al.¹⁹ conducted an *ex vivo* study involving 14 patients using texture feature extraction methods (GLCM, LM, CSAC, LBP) combined with machine learning classifiers (SVM, KNN, DT, RF), reporting an overall diagnostic accuracy of 94.1%, although sensitivity and specificity values were not explicitly provided. Yang et al.²⁰ analyzed 19 patients using OCT enhanced with deep learning (CNN LeNet-5), reporting a sensitivity of 86.0%, specificity of 97.3%, and accuracy of up to 96.8%, depending on classifier integration.

Overall, OCT demonstrated sensitivity values ranging from 81.5% to 100%, specificity values from 68.8% to 100%, and diagnostic accuracies as high as 100% in intraoperative applications. Studies incorporating machine

learning or deep learning techniques consistently reported superior diagnostic performance compared with conventional OCT interpretation.

Reported confidence intervals were available only in the studies by Sunny et al.¹⁷ and James et al.¹⁸ Due to incomplete availability of raw diagnostic data, construction of forest plots or receiver operating characteristic curves was not feasible. Nevertheless, the cumulative evidence indicates robust diagnostic accuracy of OCT for early oral cancer detection. Diagnostic accuracy outcomes for the seven included studies are summarized in Table 3.

Discussion

OCT has increasingly emerged as a highly promising, non-invasive imaging modality for the early diagnosis and prognostic assessment of oral cancer and precancerous conditions.²¹ This systematic review synthesized evidence from 7 studies, collectively demonstrating that OCT, particularly when augmented with advanced imaging approaches and sophisticated machine learning algorithms, substantially improves diagnostic accuracy beyond conventional clinical and radiographic methods. The theoretical basis of OCT's diagnostic utility lies in the principles of low-coherence interferometry and high-resolution cross-sectional imaging, which enable real-time visualization of tissue microstructure at near-histological resolution.²² This capability is critical for identifying subtle epithelial and subepithelial alterations indicative of dysplasia or malignancy, which are frequently difficult to detect using traditional visual inspection or palpation-based assessments.²³

Heidari et al.¹⁵ reported that OCT significantly outperformed conventional visual inspection scoring in resource-limited clinical environments, highlighting its potential utility for early oral cancer detection in settings where advanced diagnostic infrastructure is unavailable. This observation aligns with diagnostic accuracy theory, which emphasizes the importance of high sensitivity and specificity in screening tools to minimize false-negative and false-positive results, thereby facilitating timely and appropriate clinical intervention. Similarly, Wessels et al.²⁴ demonstrated that OCT can non-invasively detect epithelial alterations characteristic of premalignant and early malignant lesions with high sensitivity, further supporting the role of OCT in enhancing tissue characterization beyond gross morphological assessment.

OCT provides real-time, microscopic-level visualization of tissue microarchitecture, enabling identification of architectural and morphological alterations at the cellular level

Table 3. Summary of recent studies using optical coherence tomography (OCT) for oral cancer detection

Author, year	Country	Sample	Study design	Diagnostic assessment		Sensitivity (%)	Specificity (%)	Accuracy (%)
				OCT with or without AI	Conventional			
Hamdoon et al. ¹⁴ 2016	UK	20 patients	Prospective <i>ex vivo</i> diagnostic accuracy study	OCT	Histopathology	81.5	87	84
Heidari et al. ¹⁵ 2019	USA	28 patients	Prospective clinical <i>in vivo</i> diagnostic accuracy study	OCT	Clinical examination, histopathology	OCT: 95 Conventional: 89	OCT: 100 Conventional: 100	N/A
Jerjes et al. ¹⁶ 2019	UK	60 patients	Prospective <i>ex vivo</i> diagnostic accuracy study	OCT (measurement of epithelial thickness)	Histopathology	92.2	94.2	N/A
Sunny et al. ¹⁷ 2019	India	14 patients	Prospective <i>in vivo</i> intraoperative diagnostic accuracy study	OCT	Histopathology: malignancy Histopathology: dysplasia	100 (95%; CI: 85.2-100) 92.5 (95%; CI: 84.4-97.2)	100 (95%; CI: 85.8-100) 68.8 (95%; CI: 41.3-88.9)	100 88.5
James et al. ¹⁸ 2021	India	232 patients	Prospective <i>in vivo</i> intraoperative diagnostic accuracy study	OCT + simple ML OCT + ANN-SVM	Histopathology	93 (95% CI: 82-98) 86	74 (95% CI: 65-82) 81	85
Yang et al. ¹⁹ 2021	China	14 patients	<i>Ex vivo</i> diagnostic accuracy study	OCT with texture feature (GLCM, LM, CSAC, LBP) + ML classifiers (SVM, KNN, DT, RF)	Histopathology	N/A	N/A	94.1
Yang et al. ²⁰ 2023	China	19 patients	<i>Ex vivo</i> diagnostic accuracy study	OCT CNN (LeNet5) + ML (SVM)	Histopathology	86.0	97.3	92.52

AI: artificial intelligence, N/A: not applicable, ML: machine learning, CNN: convolutional neural network, ANN: artificial neural network, GLCM: gray level co-occurrence matrix, LM: Laws' texture measures, CSAC: center symmetric auto-correlation, LBP: local binary pattern, SVM: support vector machine, KNN: K-nearest neighbor, DT: decision tree, RF: random forest

that are often imperceptible with conventional imaging techniques. The primary advantages of OCT include its capacity to perform rapid, painless examinations without the need for invasive tissue sampling, such as biopsy, making it particularly suitable for early detection, diagnosis, and longitudinal monitoring of disease processes, including oral cancer. In the context of oral oncology, OCT offers valuable information regarding epithelial thickness, basal layer disruption, and early structural abnormalities that may signal malignant transformation. With three-dimensional imaging capability and micron-scale spatial resolution, OCT can effectively differentiate between healthy and pathological tissue.^{21,23}

In addition, OCT facilitates rapid and accurate intraoperative assessment of surgical margins, assisting surgeons in achieving complete tumor excision while minimizing the risk of residual disease. Beyond diagnostic applications, OCT contributes meaningfully to biomedical and clinical research by enabling longitudinal visualization of tissue changes without tissue destruction, thereby supporting monitoring of treatment response and individualized therapeutic planning. Furthermore, OCT integrates effectively with complementary imaging modalities, including fluorescence imaging and digital histology, enhancing the overall depth and clinical utility of diagnostic and prognostic information. The incorporation of AI, particularly deep learning frameworks, further amplifies OCT's diagnostic capability by automating feature extraction and image classification processes. Yang et al.²⁰ reported a diagnostic accuracy of 96.76% for oral cancer detection using convolutional neural networks applied to OCT images, outperforming conventional machine learning models. This improvement is consistent with machine learning theory, which explains how hierarchical feature learning and non-linear pattern recognition reduce observer-dependent variability and improve diagnostic consistency across heterogeneous datasets.^{20,24}

Subasi et al.²⁵ further emphasized the transformative role of AI in medical imaging diagnostics, underscoring its ability to reduce subjective bias and enhance reproducibility, both of which are critical to clinical decision-making and patient management.²⁵ Beyond lesion detection, OCT's capacity to quantitatively assess epithelial thickness with high precision was demonstrated by Jerjes et al.,¹⁶ who reported sensitivity and specificity values of 92% and 94%, respectively, for delineating cancer margins. This quantitative capability is particularly important for surgical planning and complete tumor resection, aligning closely with precision medicine principles that advocate for individualized inter-

ventions based on patient-specific anatomical and pathological characteristics.

James et al.¹⁸ validated AI-enhanced point-of-care OCT systems, reporting sensitivities and specificities exceeding 90% for the detection of potentially malignant and malignant oral lesions. These findings underscore the feasibility of deploying rapid, accurate screening tools across diverse clinical environments. This application highlights the translational potential of OCT combined with AI, bridging the gap between advanced diagnostic technologies and real-world clinical practice.

Despite the favorable diagnostic performance of OCT demonstrated in this systematic review, several limitations warrant consideration. The included studies exhibited substantial heterogeneity in study design, sample size, imaging protocols, and machine learning configurations, which limits the generalizability of findings across broader clinical contexts. Moreover, the predominance of cross-sectional and retrospective designs restricts insight into longitudinal lesion progression and treatment outcomes. The lack of standardized OCT acquisition parameters and AI integration frameworks further impedes reproducibility and multi-center validation. Future research should therefore prioritize large-scale, prospective cohort studies employing harmonized imaging protocols, standardized AI pipelines, and rigorous external validation. In addition, inclusion of diverse patient populations and evaluation under real-time clinical deployment conditions will be essential to establish OCT's diagnostic robustness and facilitate its integration into routine oral oncology practice.

OCT, particularly when enhanced by deep learning techniques, represents a highly accurate and non-invasive modality for the detection and prognostic assessment of oral cancer and precancerous lesions. When combined with systemic biomarkers and AI, OCT has the potential to further refine diagnostic precision and optimize treatment planning, positioning it as a valuable adjunct in oral oncology. Nevertheless, large-scale, prospective, and methodologically standardized investigations remain necessary to confirm its clinical utility and support widespread implementation, with the ultimate goal of improving early diagnosis and patient outcomes.

Conflicts of Interest: None

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